

Portfolio

Data Analyst

Google Certified

Maria Aslam

PortFolio

Hi, I am Maria Aslam

A Data Analyst & BI Developer

I am a Google Certified Professional Data Analyst & Certified BI Developer focused on turning complex data into meaningful insights that drive informed decision-making.

About Me

Personal Info

First name: Maria Nationality: Pakistani
Last name: Aslam City: Lahore

Certificates

- Professional Data Analytics
By Google
- Data Analytics & Business Intelligence
By DigiSkills

Education

- Bachelor’s in Physics | 2021 -2025
Lahore College for Women University, Lahore

Skills & Tool

Data & Analytics Tools

- Power BI
- Tableau
- Excel
- Google Sheets,

Programming & Scripting

- R (tidyverse, dplyr, ggplot2, readr)
- Python (Pandas, NumPy, Matplotlib, Seaborn, Plotly)

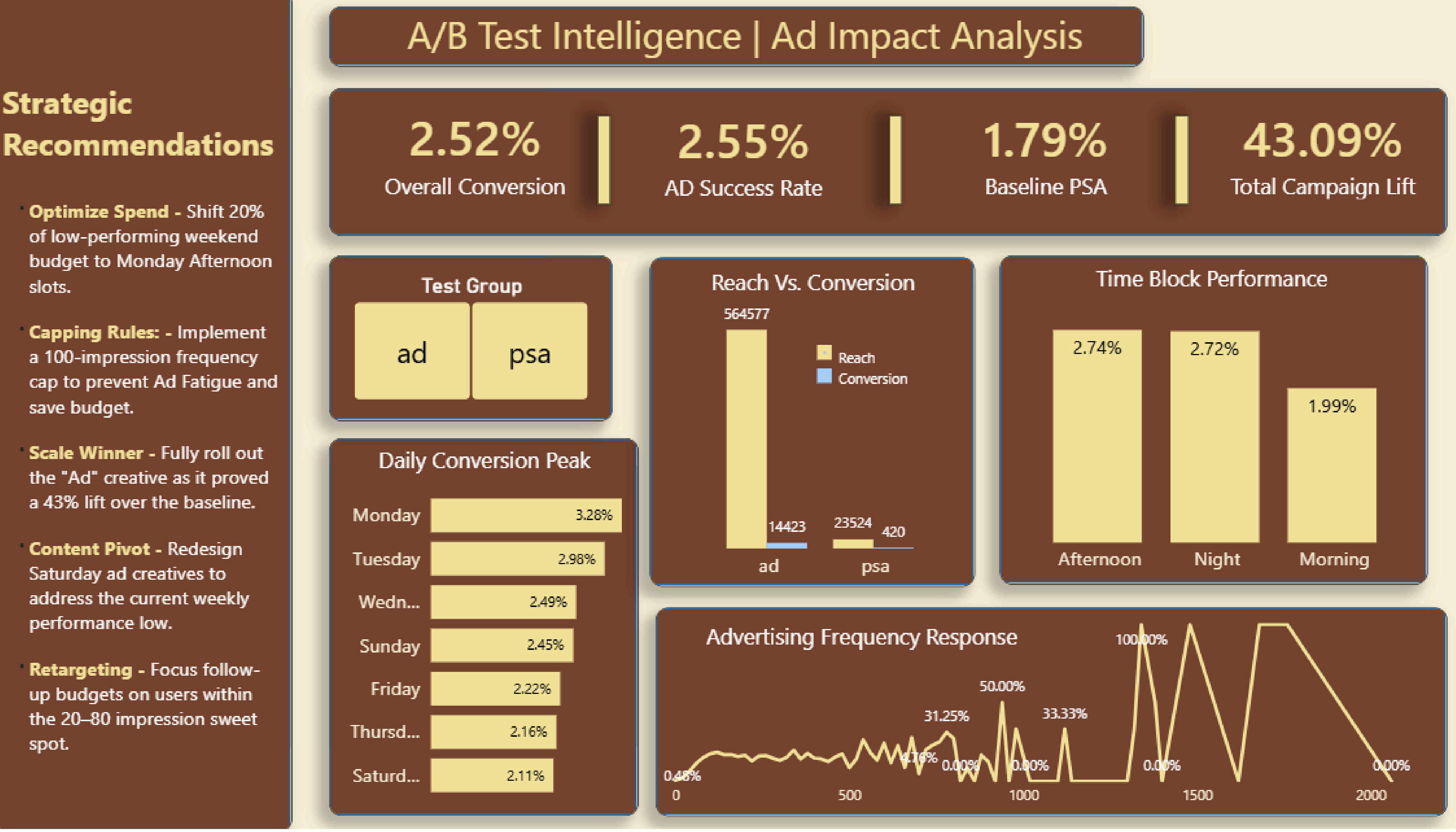
Database & Querying

- SQL (BigQuery)

Experience

- Freelance Data Analyst | Dec 2025 Present**
Worked on end-to-end data analysis projects. Performed data cleaning, analysis, & dashboard creation. Delivered insights to support data-driven decisions and performance improvement
- Teaching Assistant & Data Support | Part time | Jan 2023 - Jan 2024**
Maintained & organized financial and administrative records using MS Excel. Handled and structured data to improve accuracy & reporting efficiency. Supported data tracking and record management in an academic environment.

PROJECTS



Bank Customer Churn Analysis Report

Maria Aslam

2026-01-26

Executive Summary

Strategic Overview

This report analyzes customer attrition across our European banking operations. While France and Spain maintain stable churn rates (~16%), the bank faces a critical emergency in the German market, where churn has spiked to 32.4%. This analysis shifts focus from *who is leaving* to *why our most valuable segments are exiting*.

Key Findings

- **TheTwo-Product Anchor:** We have identified a Survival Sweet Spot where customers with exactly 2 products have the highest loyalty (11–13% churn).
- **Product Over-Saturation:** A catastrophic failure occurs when German customers reach 3 or 4 products, resulting in a 90–100% exit rate. This suggests our complex bundles are driving customers away rather than locking them in.
- **Premium Segment Flight:** Churn is not limited to low-value accounts. High-balance customers and those in the “Very Good” credit tier (~33.5% churn) are leaving at nearly the same rate as high-risk segments.
- **Service Failure:** There is a near-total correlation between complaints and attrition. Currently, a complaint is a “guaranteed exit signal,” indicating our resolution process is failing to recover dissatisfied users.

Core Recommendations

- **Plus-One Retention:**Focusmarketing on moving 1-product holders to 2 products; avoid pushing to 3+ until the *Product Fatigue* issue is resolved.
- **German Priority Desk:** Launch a localized retention program for high-credit/high-balance German users to counter aggressive competitor poaching.
- **Active Recovery:** Implement a proactive “Re-engagement” protocol for inactive German users before they hit the 41.1% churn threshold

Data Cleaning & Methodology

Toensure theintegrity of thisanalysis, theraw dataset underwent arigorouspreprocessing phase within the R environment. This ensures that all statistical summaries and visualizations reflect accurate customer behavior.

The Cleaning Process

- **DimensionalityReduction:** Irrelevant features such as RowNumber, CustomerId, and Surname were removed. These variables contain unique identifiers that do not contribute to churn patterns and would introduce “noise” into the analysis.
- **Attribute Re-classification:** Several columns were imported as integers but represent categorical states. I converted Exited, HasCrCard, IsActiveMember, and Complain into Factors. This allows the analysis to treat them as distinct groups rather than continuous numerical values.
- **Data Consistency Check:** I verified the Geography column for missing values and spelling inconsistencies to ensure the reliability of the regional “Deep Dive” results.
- **Metric Calculation Logic:** During the summarization of churn rates, a nested as.numeric(as.character()) transformation was used. This ensures that binary factor levels are correctly interpreted for accurate percentage calculations.

Detail Analysis

Thedatasetcontains customer demographic, financial, and behavioral information used to analyze churn patterns. **Total customers ≈ 10,000**

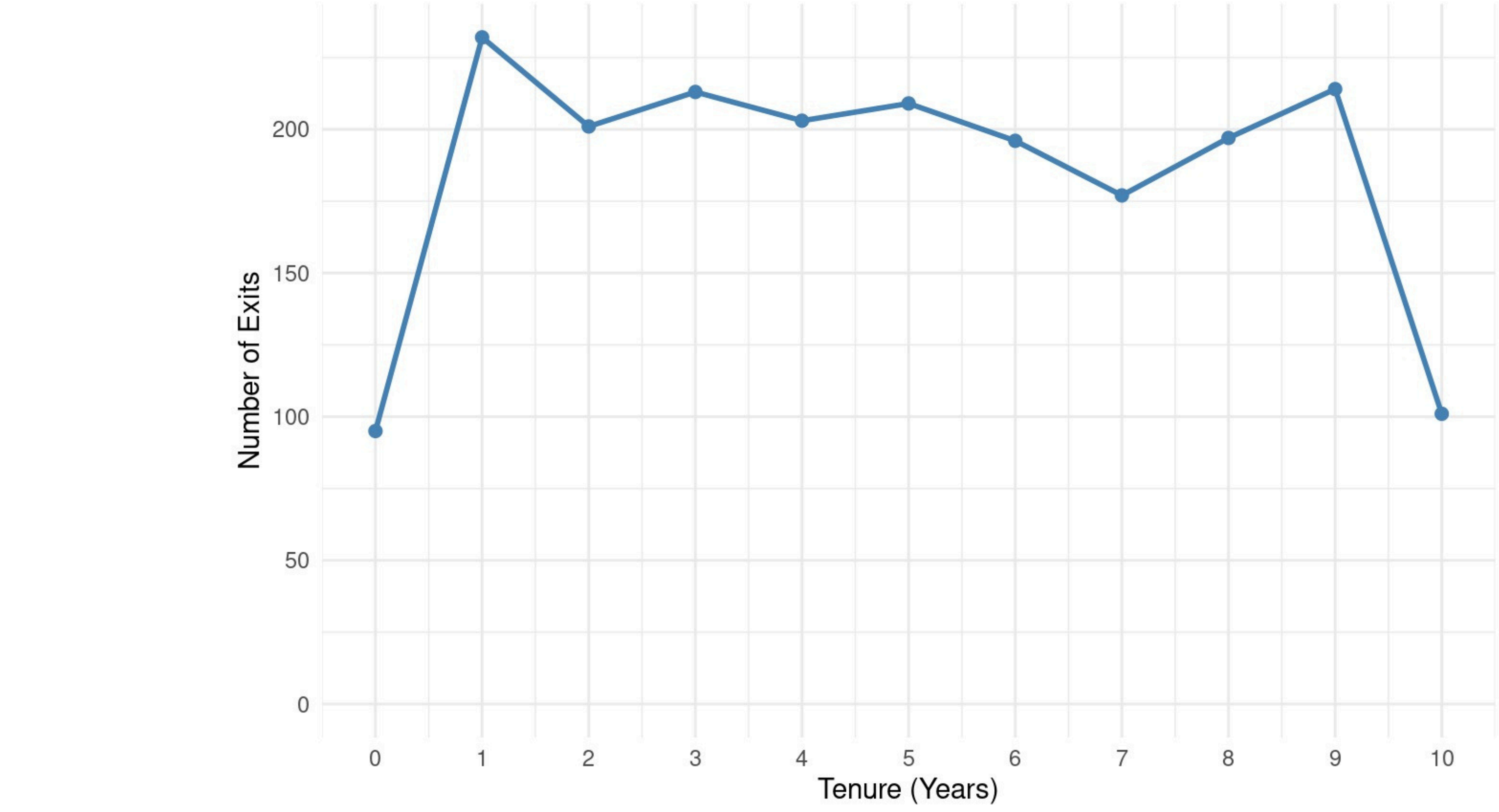
- Countries:**
- France
 - Germany
 - Spain

Quick Summary

Total Customers	Total Exited	Active Members	Total Complaints	Churn Rate
10000	2038	5151	2044	20%

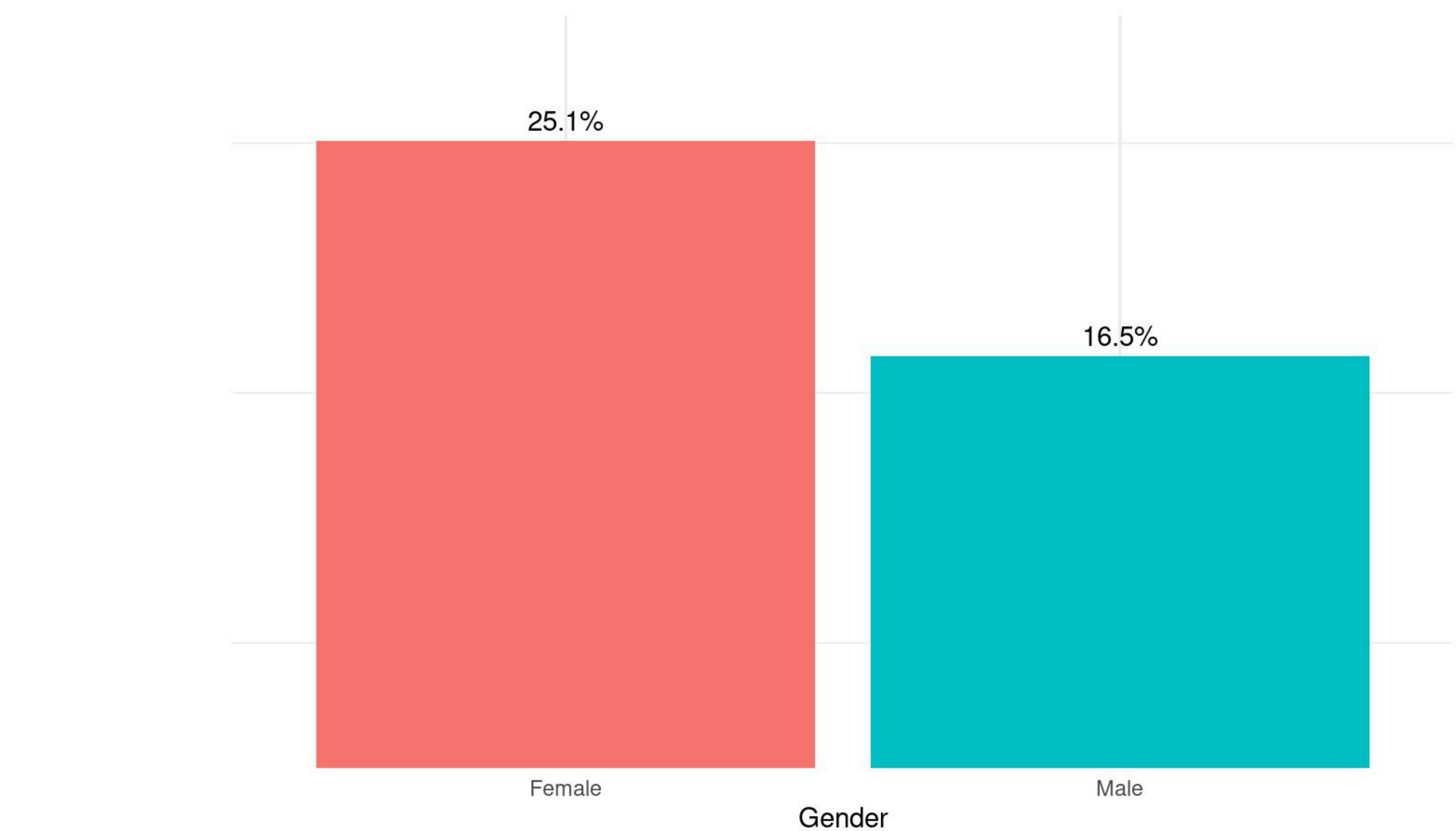
Approximately **20% of customers have churned**,indicating moderate customer attrition risk.

Tenure Analysis



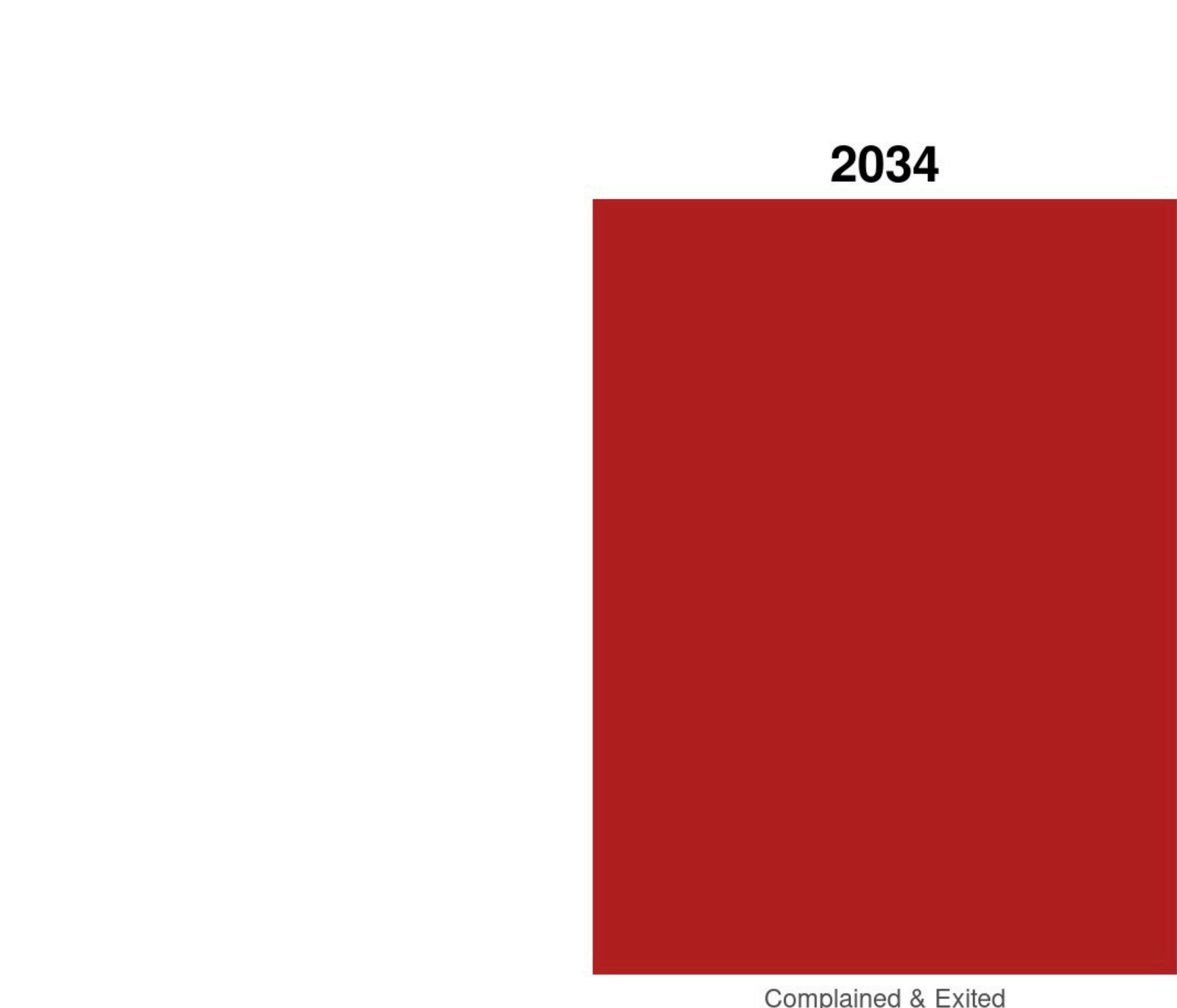
The data contradicts the standard **loyalty curve** where churn should decrease as tenure increases. To improve retention, the focus must shift from just winning back new joiners to addressing the **sustained dissatisfaction** that causes veteran customers of 5+ years to leave at the same rate as those of 0-2 years.

Churn by Gender



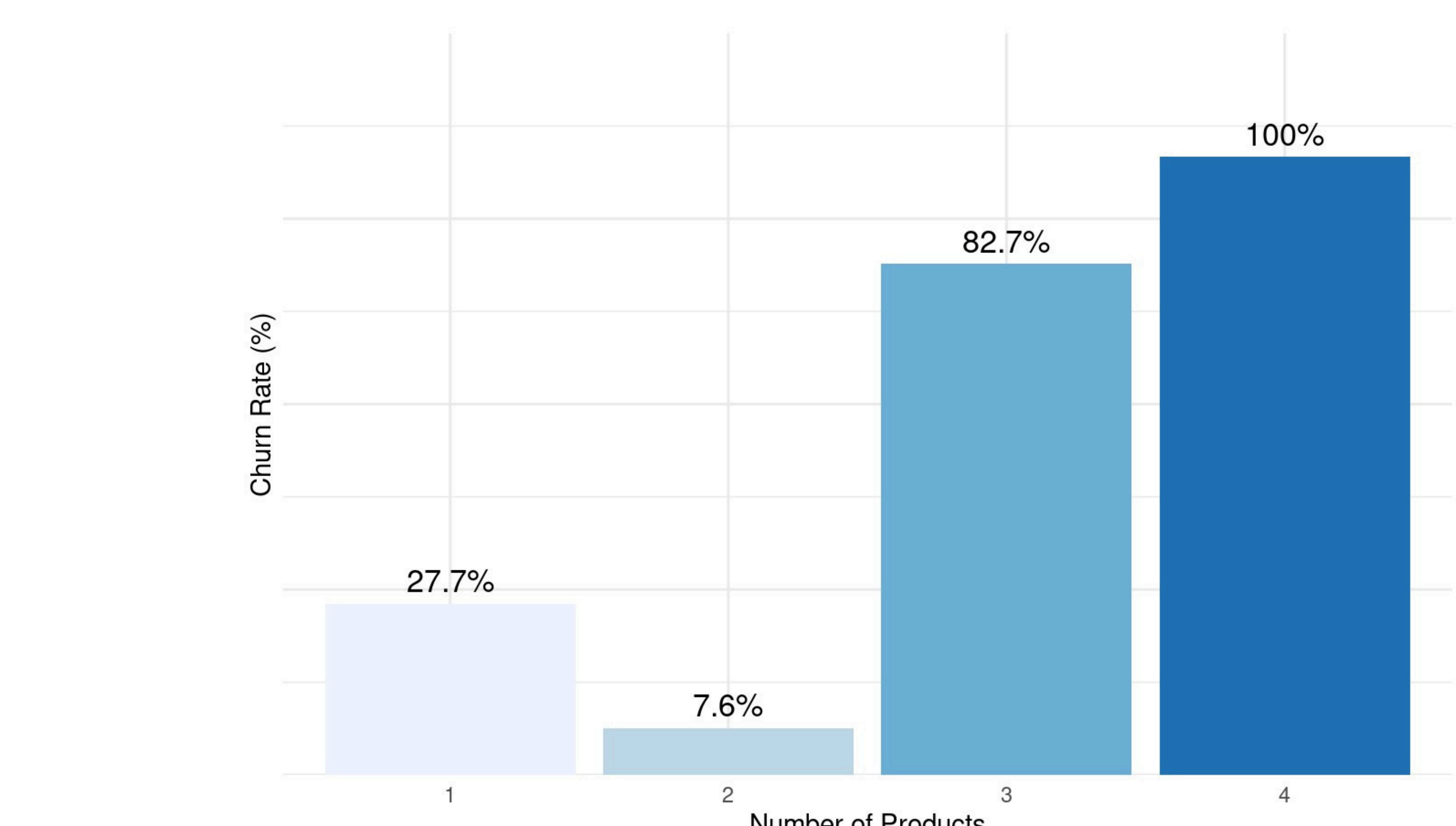
There is a stark contrast in retention across gender lines: Female customers exhibit a 25.1% churn rate, while Male customers churn at 16.5%. This represents an **8.6% absolute difference**, meaning women are roughly **1.5 times** more likely to leave the bank than men.

Complain Impact On Churn



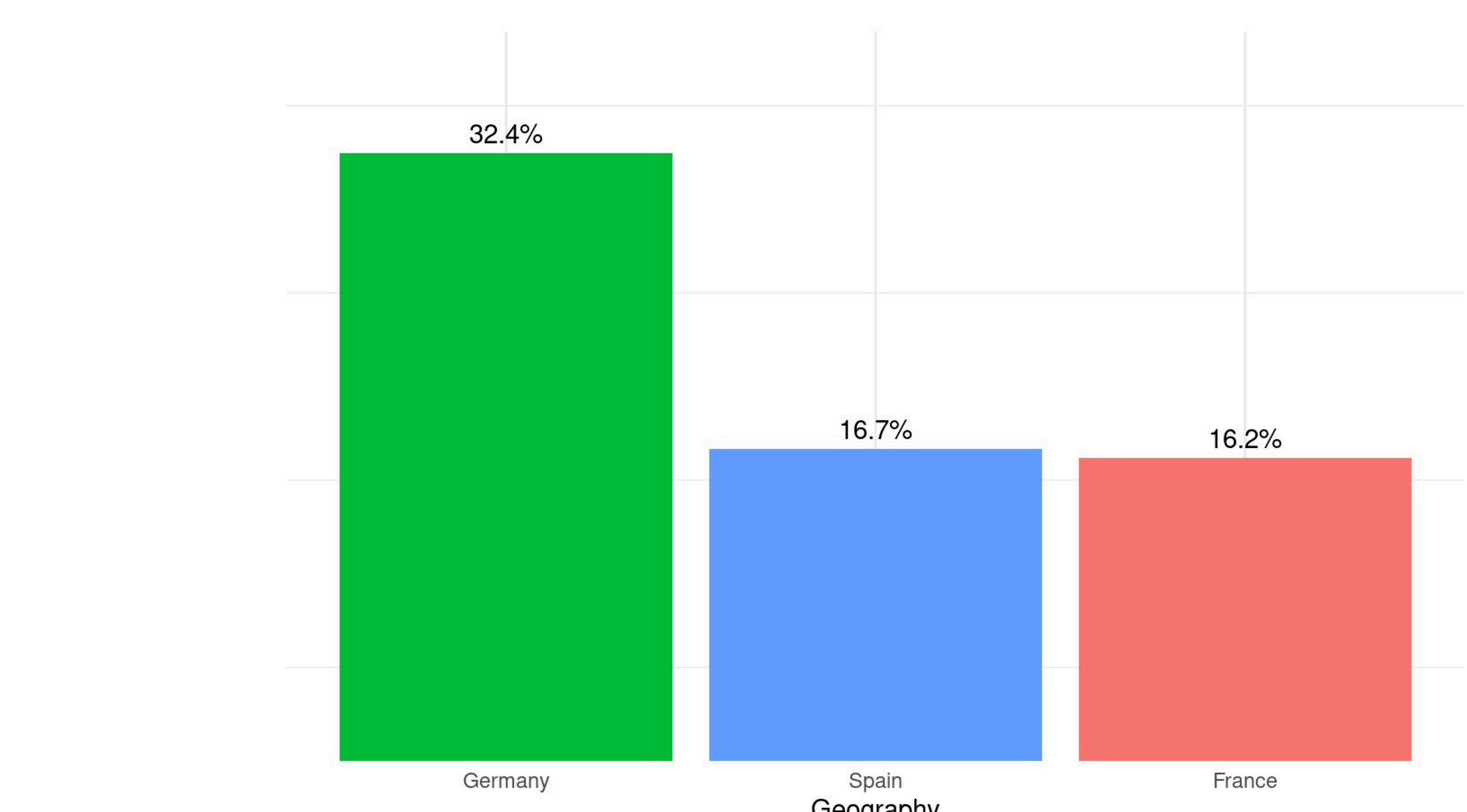
2,034 customers transitioned from Complaint Filed to Account Closed, representing a **100% failure** rate in complaint-driven retention

Product Vs Churn



The data reveals a *U-Shaped risk curve*. Customers with 2 products have the highest loyalty (lowest churn). However, churn increases significantly for Single-Product customers and **spikes dangerously** for those with 3 or 4 products (often exceeding (80-90%))

Churn by Geography

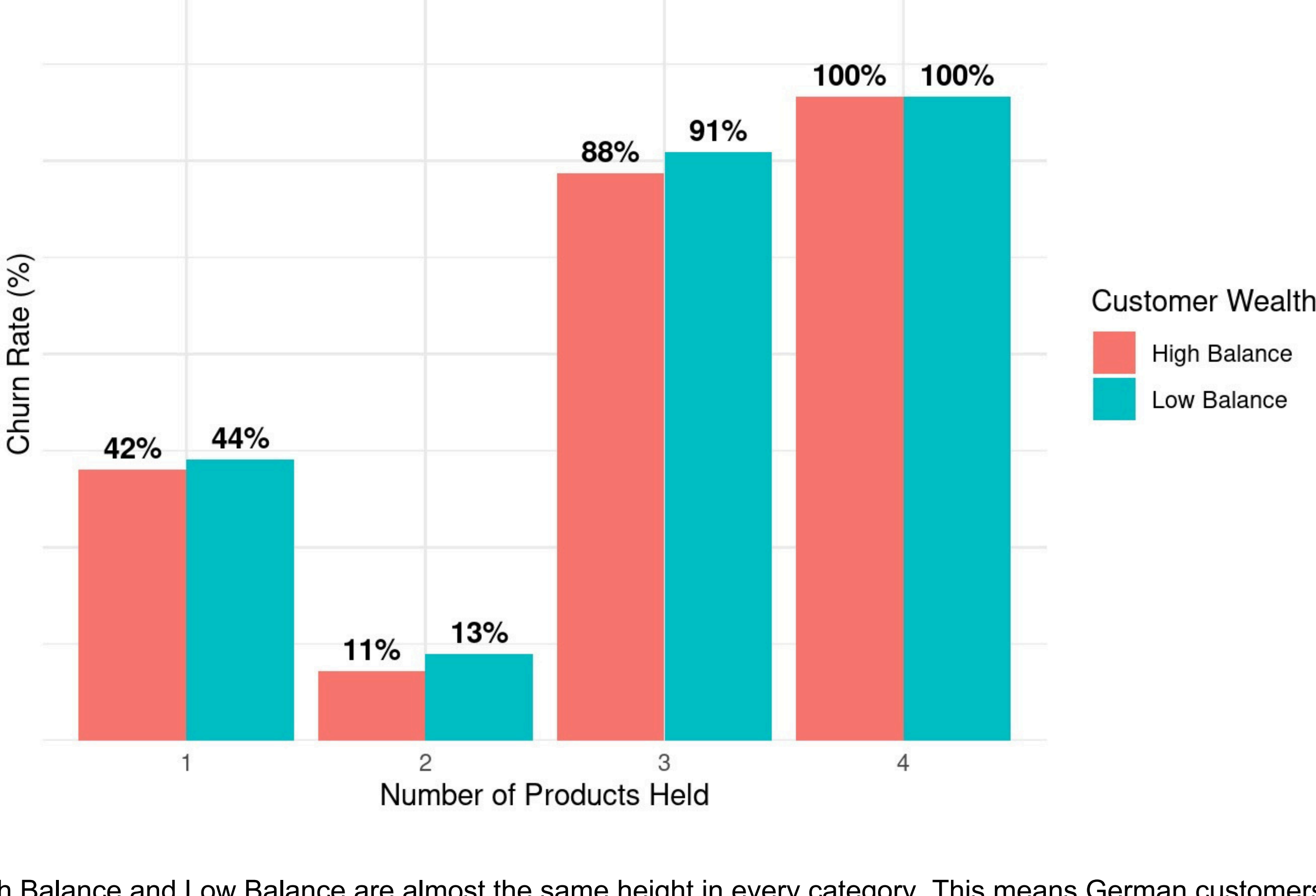


Germany requires an immediate **Deep Dive Audit**. We need to look at the German segment's demographics, average account balances, and product holdings to see if we are losing our most profitable customers or if this is a general exodus

Deep Dive Analysis | Germany

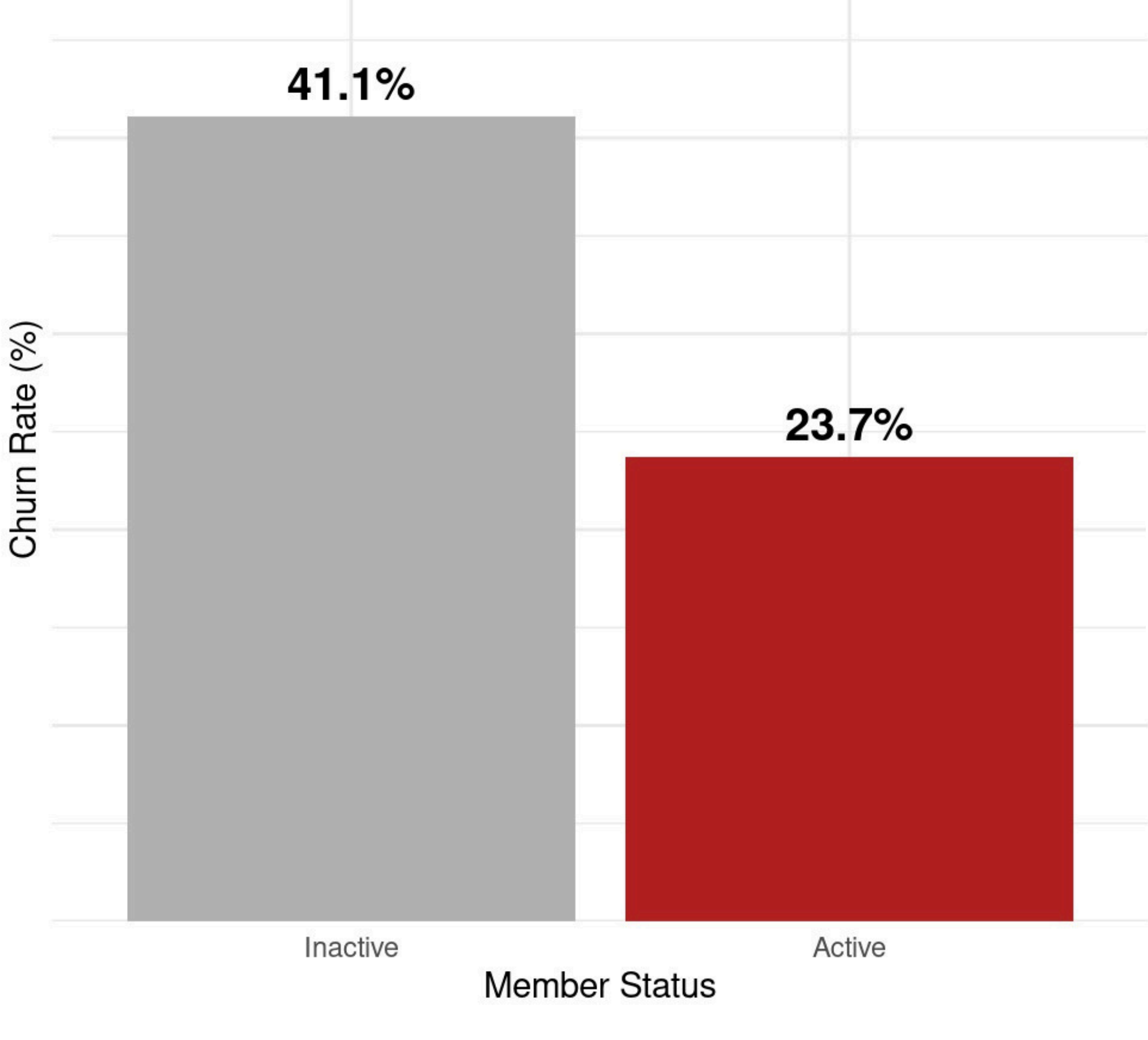
Analysis On Why is Germany's Churn Rate Significantly Higher ?

Does Account Balance Influence Churn in Germany?



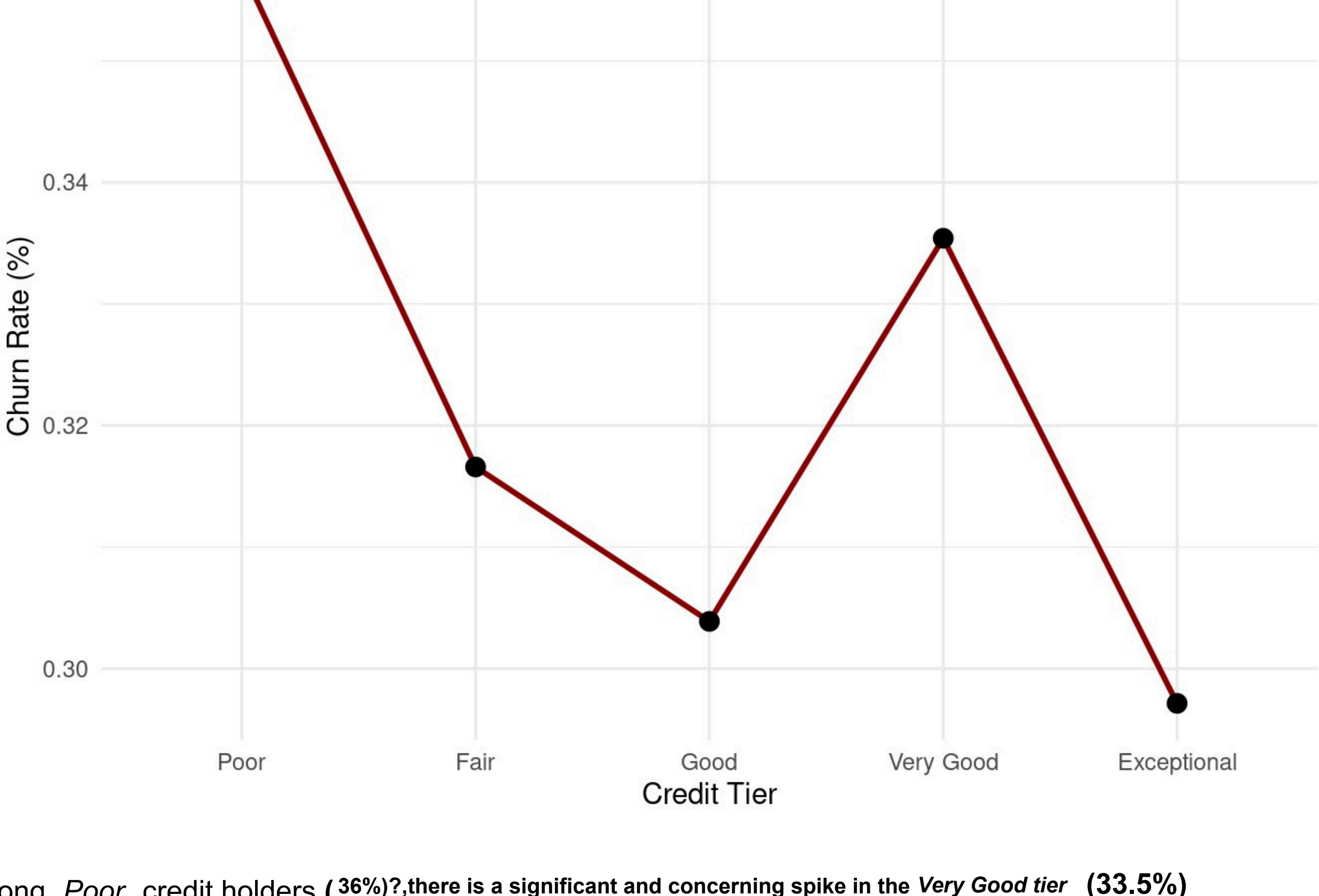
Notice that the High Balance and Low Balance are almost the same height in every category. This means German customers aren't leaving because they are rich or poor; they are leaving based on how many products they hold

Can Customer Engagement Help Reduce Churn in Germany?



Activity is a major factor in retention, but it is **not a cure** Inactive members in Germany are in a state of collapse with a 41.1% churn rate, while Active members still churn at 23.7%

How Credit Quality Affects Customer Churn in Germany?

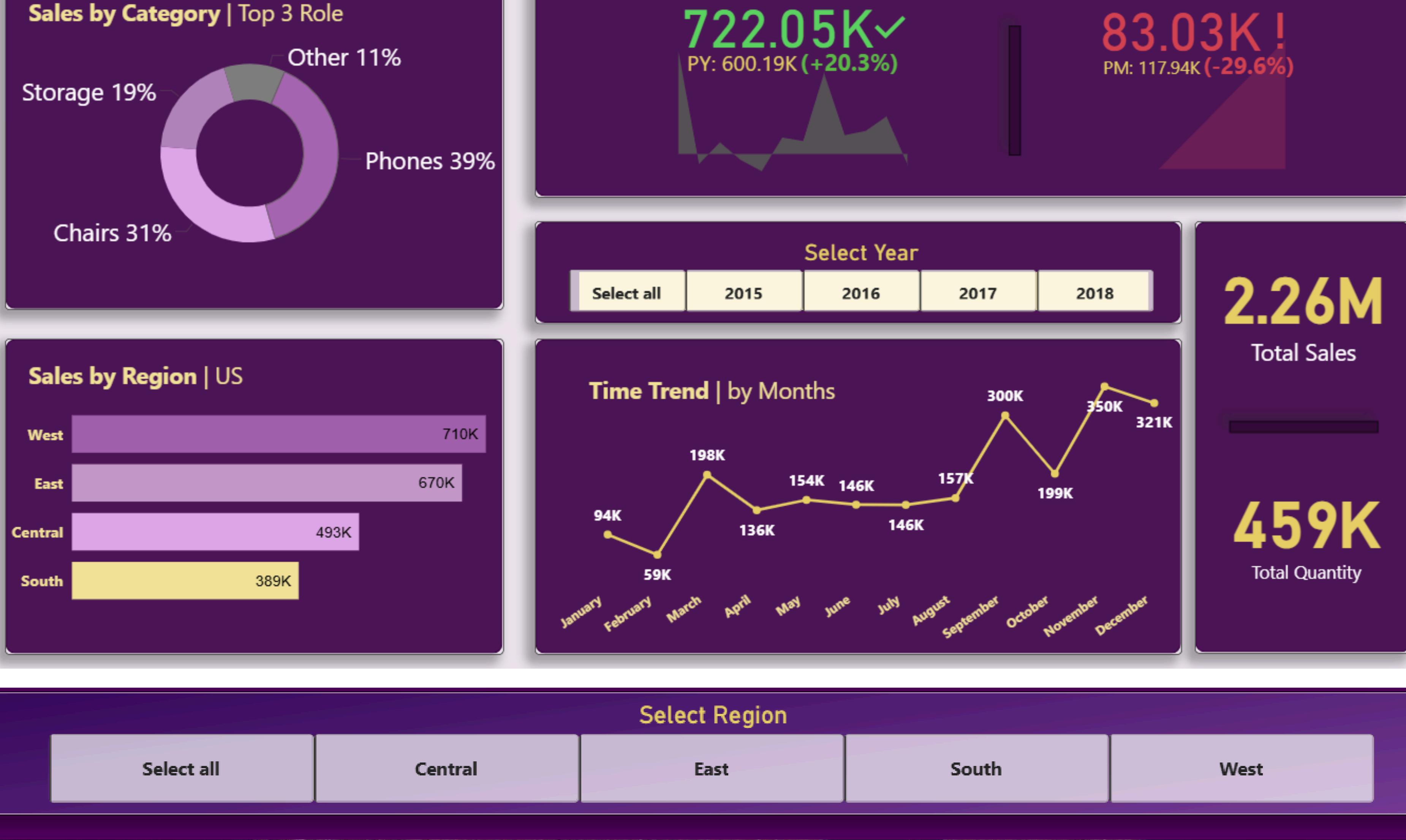


Highest churn is among *Poor* credit holders (36%)?,there is a significant and concerning spike in the *Very Good tier* (33.5%)

Strategic Conclusion

The data confirms that the bank's issue in Germany is not a lack of customer engagement, but a **gap in premium relationship management**. We are successfully attracting high-value, active, and credit-worthy customers, but our current ecosystem—specifically our complaint resolution and multi-product bundling—is creating *friction* that drives them to competitors. By stabilizing the German market through a *Plus-One* product strategy and a dedicated complaint-recovery desk, we can protect the bank's most profitable assets and reduce regional churn back to sustainable levels.

Sales at a Glance



Marketing Campaign Performance & ROI Audit

Maria Aslam

Janurary 2026

Executive Summary

Strategic Overview

This report evaluates the performance of **\$≈ 62.37 Million** in total marketing expenditure over a rolling 12-month period. While the portfolio maintains a stable **6.01% Weighted ROI**, the analysis reveals a Performance Ceiling caused by top-of-funnel engagement gaps. This summary shifts focus from general channel maintenance to creative optimization as the primary driver for future growth.

Key Findings

- The Stability Paradox:** ROI remains virtually identical across all channels and customer segments. This indicates a mature, low-risk portfolio but suggests that budget reallocation alone will not generate a performance breakthrough.
- Top-of-Funnel Bottleneck:** We are achieving massive scale in impressions, but facing a significant drop-off to Clicks (8.5% CTR). The current strategy is successful at reaching people but failing to "hook" them at the first touchpoint.
- Ad Fatigue Resilience:** Analysis of campaign duration (5–30 days) shows zero ROI decay. Our creative content is "evergreen," meaning we can run longer campaign flights without losing profitability.
- Cost-Driven Volatility:** The ROI dip observed in April was not a result of falling consumer demand, but rather minor fluctuations in acquisition costs. Our margins are more sensitive to internal spend efficiency than external market shifts.

Core Recommendations

- Creative Pivot (CTR Focus):** Shift 100% of A/B testing efforts toward *Thumb-Stopping* creative content to improve the **8.5% CTR**.
- Evergreen Deployment:** Transition to longer campaign durations (21+ days) to reduce operational overhead, as performance does not drop over time.
- Efficiency Buffer:** Implement a Cost-Cap protocol during the Q2 period to prevent the minor spend spikes that previously eroded our ROI margins.

Data Cleaning & Methodology

To ensure the integrity of this analysis, the raw marketing dataset underwent a rigorous preprocessing phase. This ensures that all statistical summaries and visualizations reflect accurate campaign performance and customer engagement metrics.

The Cleaning Process

- Duplicate & Integrity Check:** I performed a comprehensive scan for duplicate entries and verified that each campaign record was unique. This prevents the artificial inflation of revenue and cost metrics.
- Missing Value Strategy:** I audited the dataset for null values across critical columns like Revenue and Acquisition_Cost. Any inconsistencies were resolved to ensure a 100% complete data profile for the \$62.3M spend analysis.
- Type Casting & Formatting:** Converted the Date column into a datetime64 object to enable precise chronological sorting and seasonal trend analysis.
- Dimensionality Optimization:** Removed non-essential system identifiers and ensured that categorical features (such as Campaign_Type and Target_Audience) were properly encoded for grouping operations.
- Statistical Validation:** Every calculated metric (ROI, Conversion Rate, CTR) was cross-referenced against raw sums to ensure the "Weighted" averages accurately represent the true financial impact.

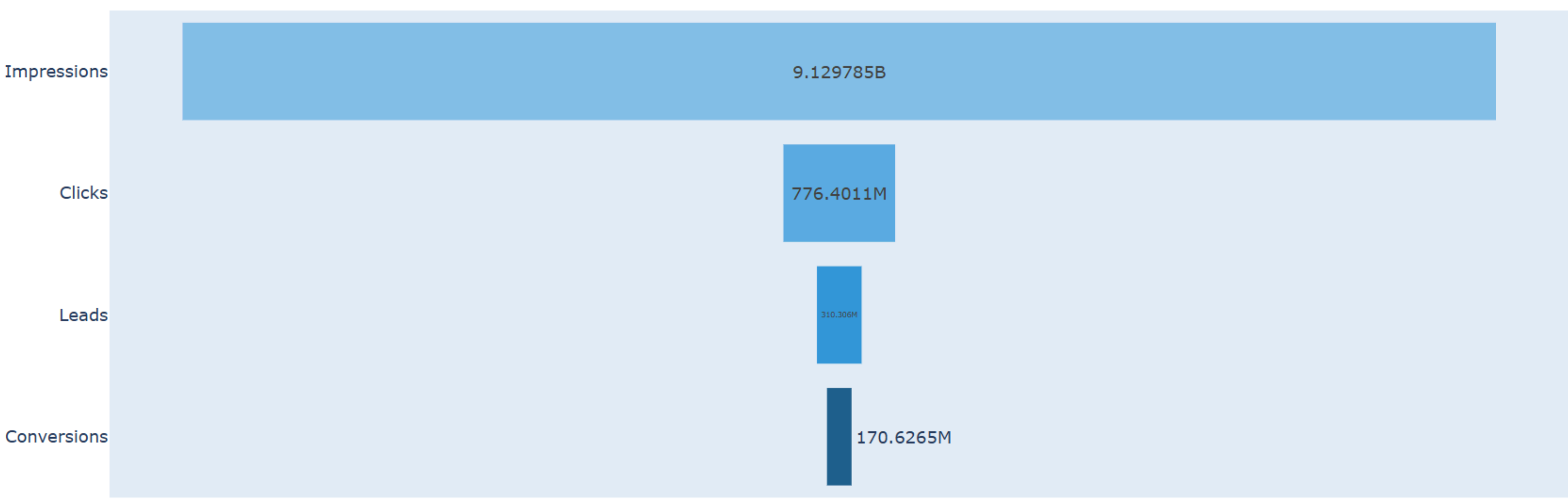
Exploratory Data Analysis - EDA

This phase involves a multi-dimensional examination of **\$62.3M** in marketing spend, cross-referencing campaign performance against localized audience behaviors and channel-specific trends. By segmenting the data through a 12-month lens, we identify the specific drivers of conversion and the mechanical bottlenecks within our sales funnel.

Quick Summary

Impressions	Clicks	Leads	Conversions	Revenue	Conv. Rate	Weighted ROI	Total Spend
9129.8M	776.4M	310.3M	170.6M	85207.5M	21.94%	6.01%	\$62,365,216.56

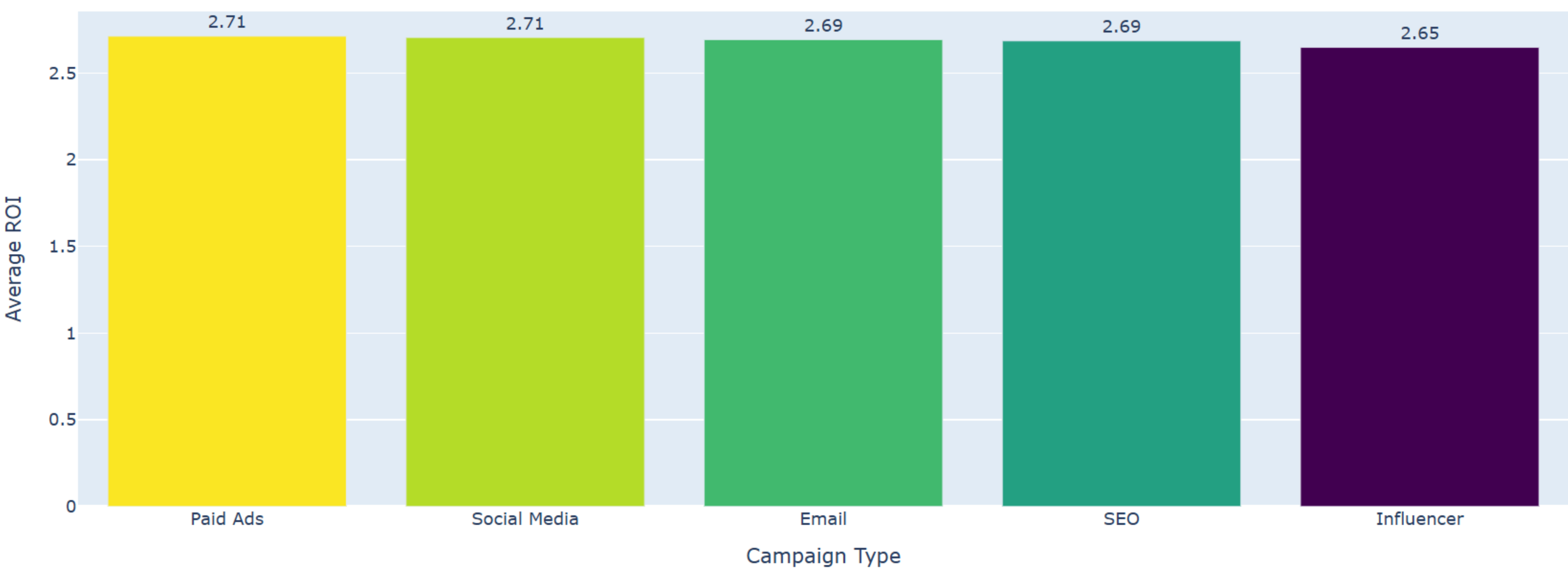
Marketing Funnel Performance



Key Takeaway: The funnel shows a significant drop-off from **Impressions 9.1B** to Clicks (776M), representing an **8.5% Click-Through Rate**. However, the conversion efficiency from Leads to Conversions remains strong, suggesting that while top-of-funnel reach is vast, audience targeting could be further refined to improve initial engagement where optimization strategies could improve customer conversion performance.

ROI Efficiency Analysis

Average ROI by Campaign Type

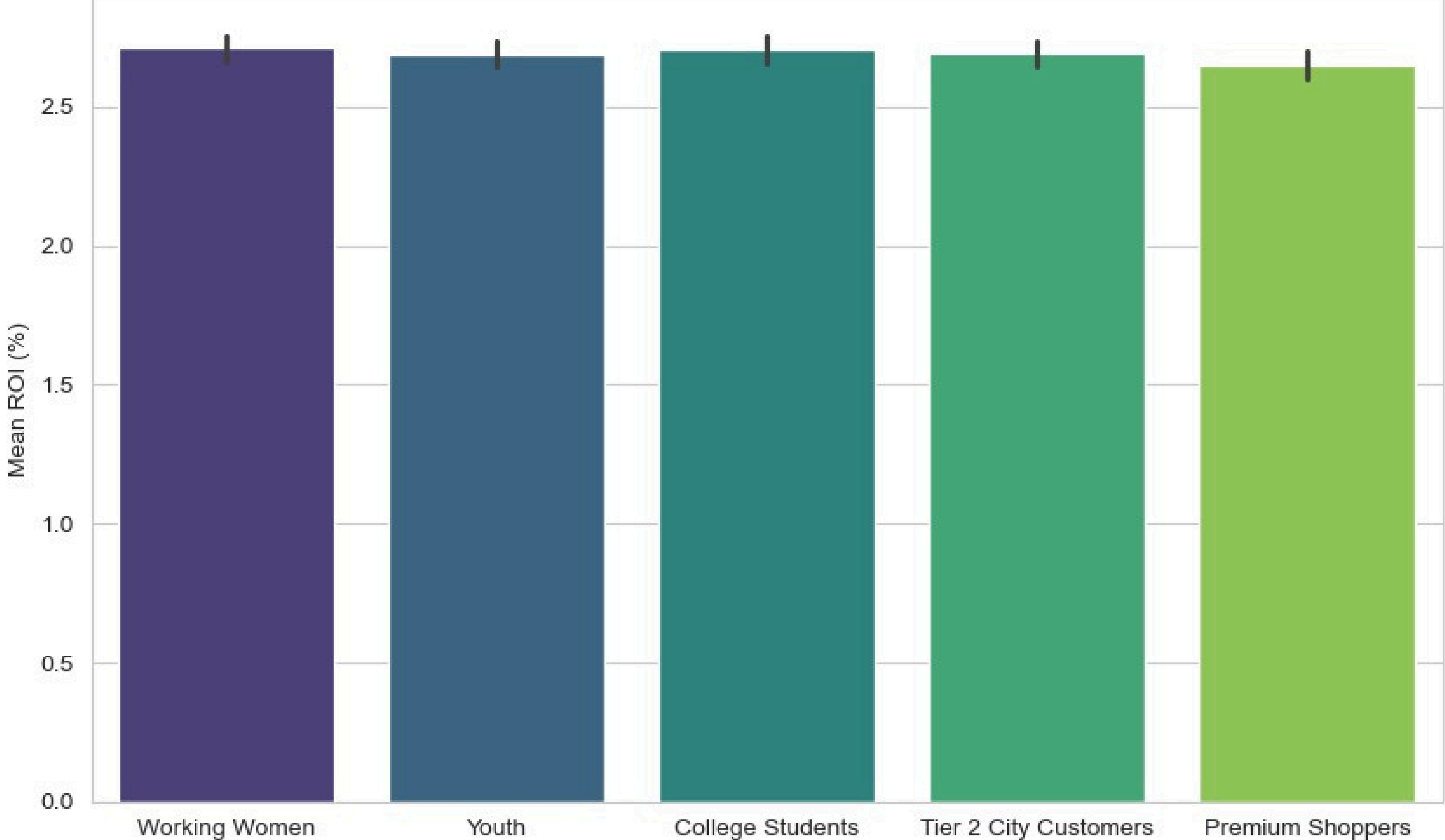


Key Takeaway: While ROI is consistent across all channels, this suggests that **efficiency is no longer the deciding factor** for budget allocation. Management should instead focus on **Volume and Scalability**, shifting funds toward channels with the highest total Revenue while maintaining this stable 6% ROI floor.

Analysis of Customer Segments

Average ROI across different customer segments to identify if specific groups respond better to our campaigns.

Average ROI by Customer Segment

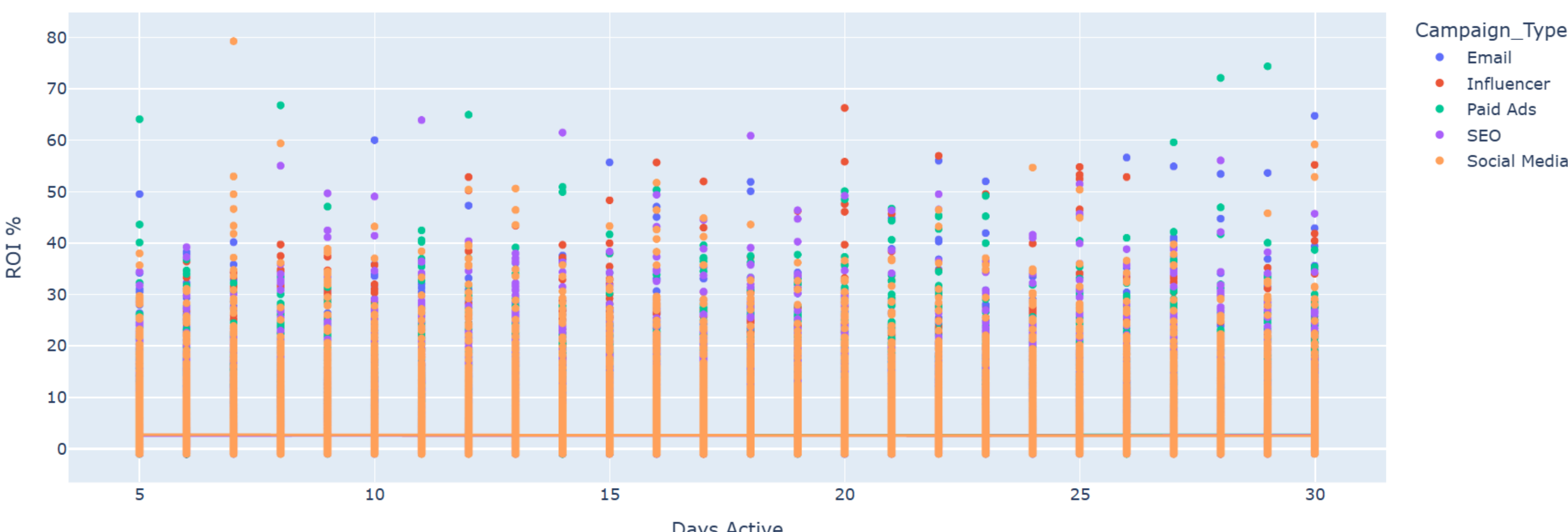


Analysis Insight: The ROI remains consistent across all customer segments, further confirming that our marketing message resonates equally across the entire database. This indicates a **highly stable and predictable revenue model**, where no single demographic is being underserved or over-exploited.

Duration Impact Analysis

We analyzed the relationship between campaign length and profitability to identify "ad fatigue."

Impact of Campaign Duration on ROI



Analysis Insight: The flat trendline in the Duration vs. ROI analysis indicates zero diminishing returns. Campaigns maintain their efficiency regardless of length (up to 30 days). This allows for longer-term campaign planning without the risk of performance decay.

ROI Trend

By looking at the full timeline from 2024 through 2025, we can see how the marketing strategy has evolved over time rather than just looking at a single calendar year.

ROI Trend

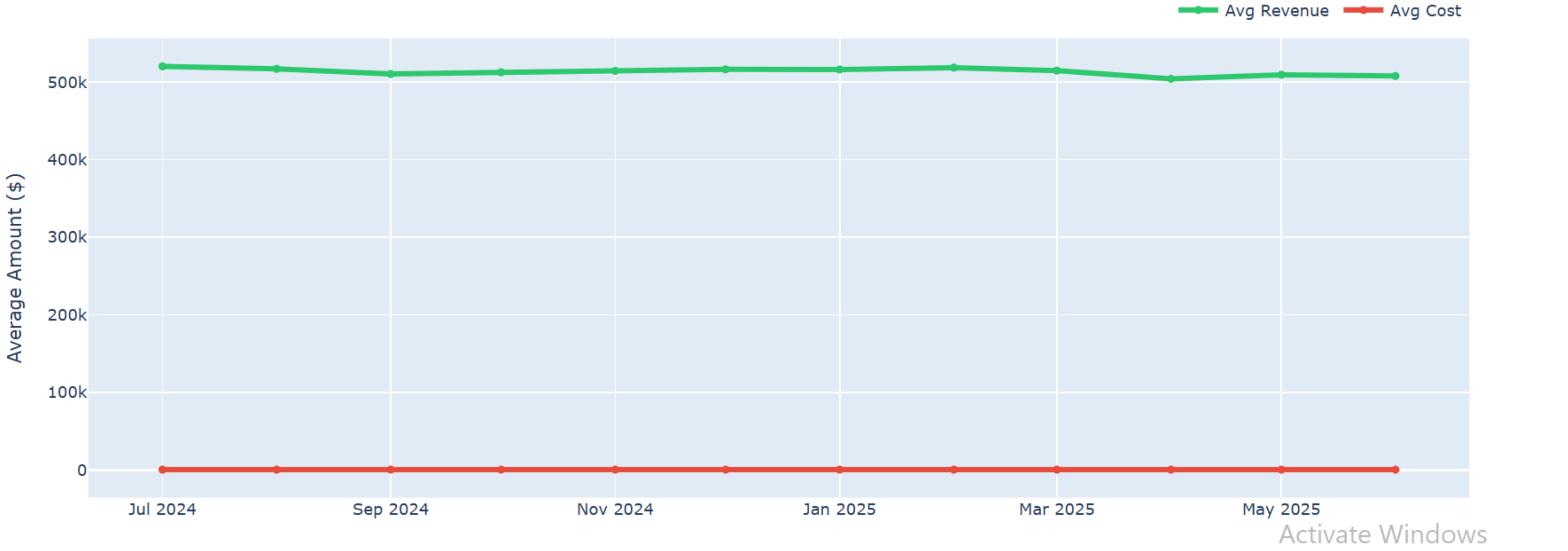


Key Takeaway: The view reveals that ROI peaked in **February 2024** and has remained relatively stable since. The lack of significant "year-over-year" growth suggests the campaign has reached a steady state of performance, providing a predictable 2.7% return on investment.

Deep Dive: Root Cause Analysis of the Q2 ROI Dip

To move beyond surface-level trends, we analyzed the direct relationship between Revenue and Acquisition Cost. This identifies whether our April performance drop was caused by rising ad costs or a decrease in consumer purchase intent.

Deep Dive: Why did ROI drop? (Revenue vs. Cost)



Root Cause Analysis: The Deep Dive chart shows that **Revenue remained stable** even during the ROI dip. This suggests that the April decline was not due to a loss of customers, but rather a slight **reduction in operational efficiency**. Since costs and revenue are so far apart, even a minor \$100 increase in cost can cause the ROI percentage to fluctuate.

Conclusion & Strategic Recommendations

Based on the multi-dimensional analysis of the marketing data from 2024-2025, we have identified three core strategic pillars for the upcoming quarter.

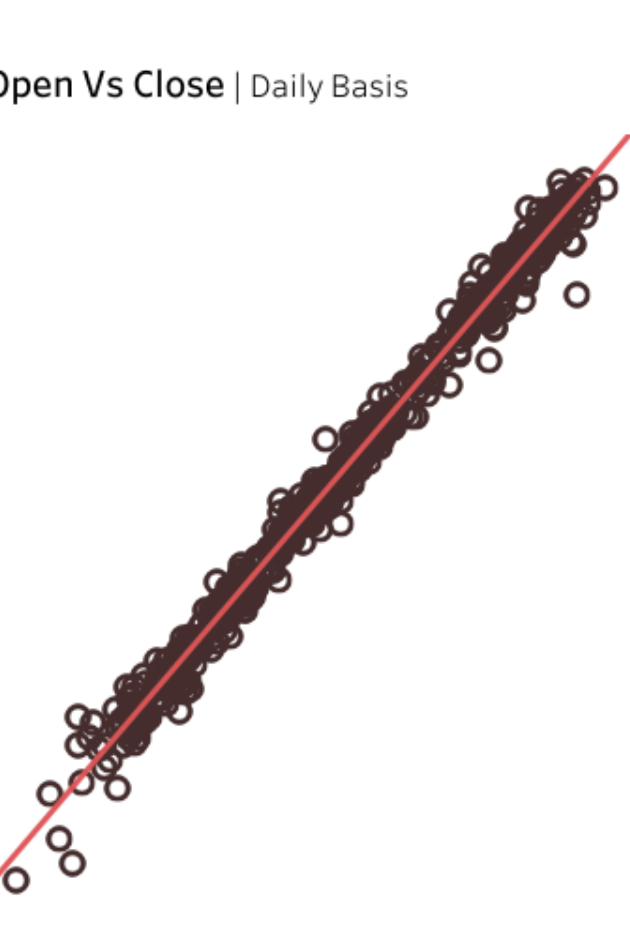
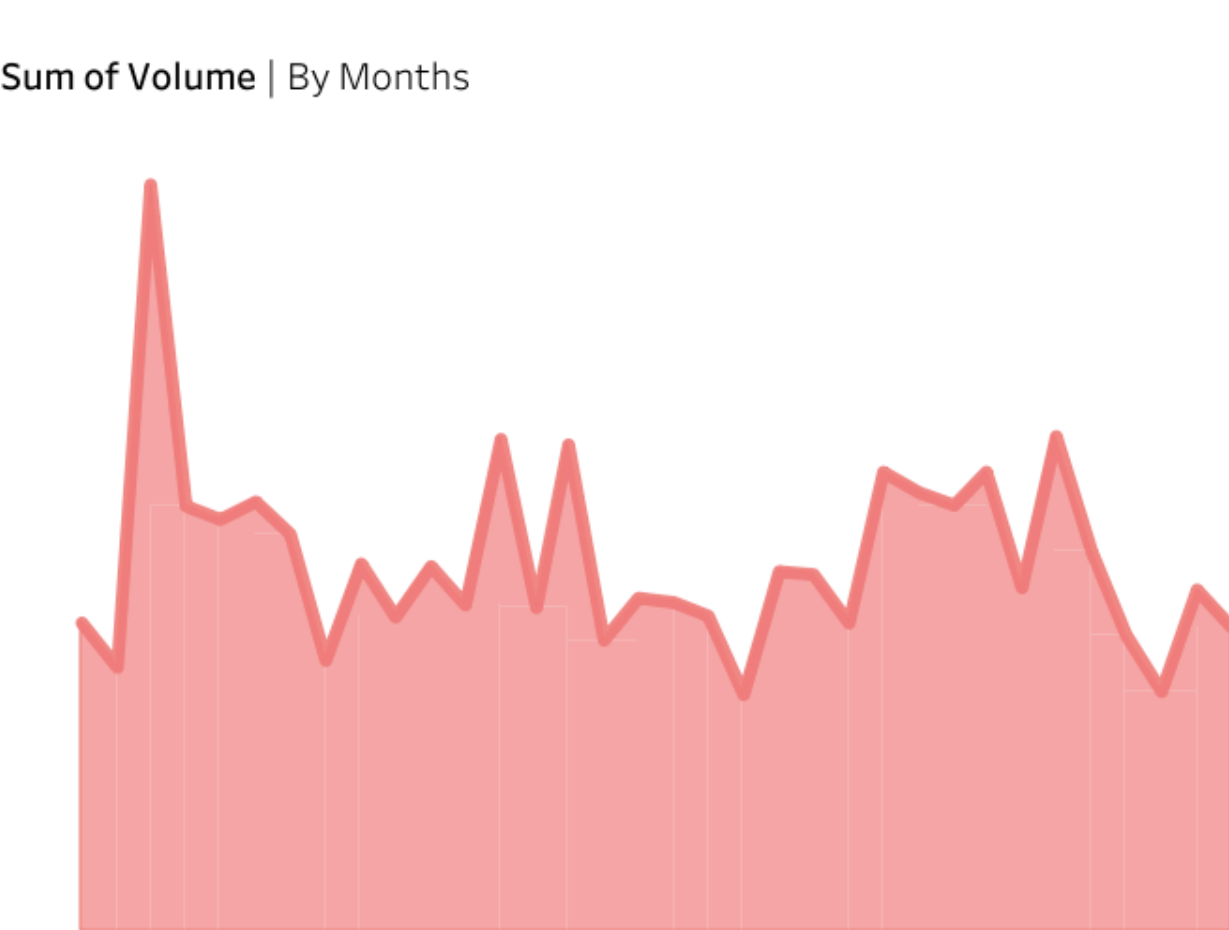
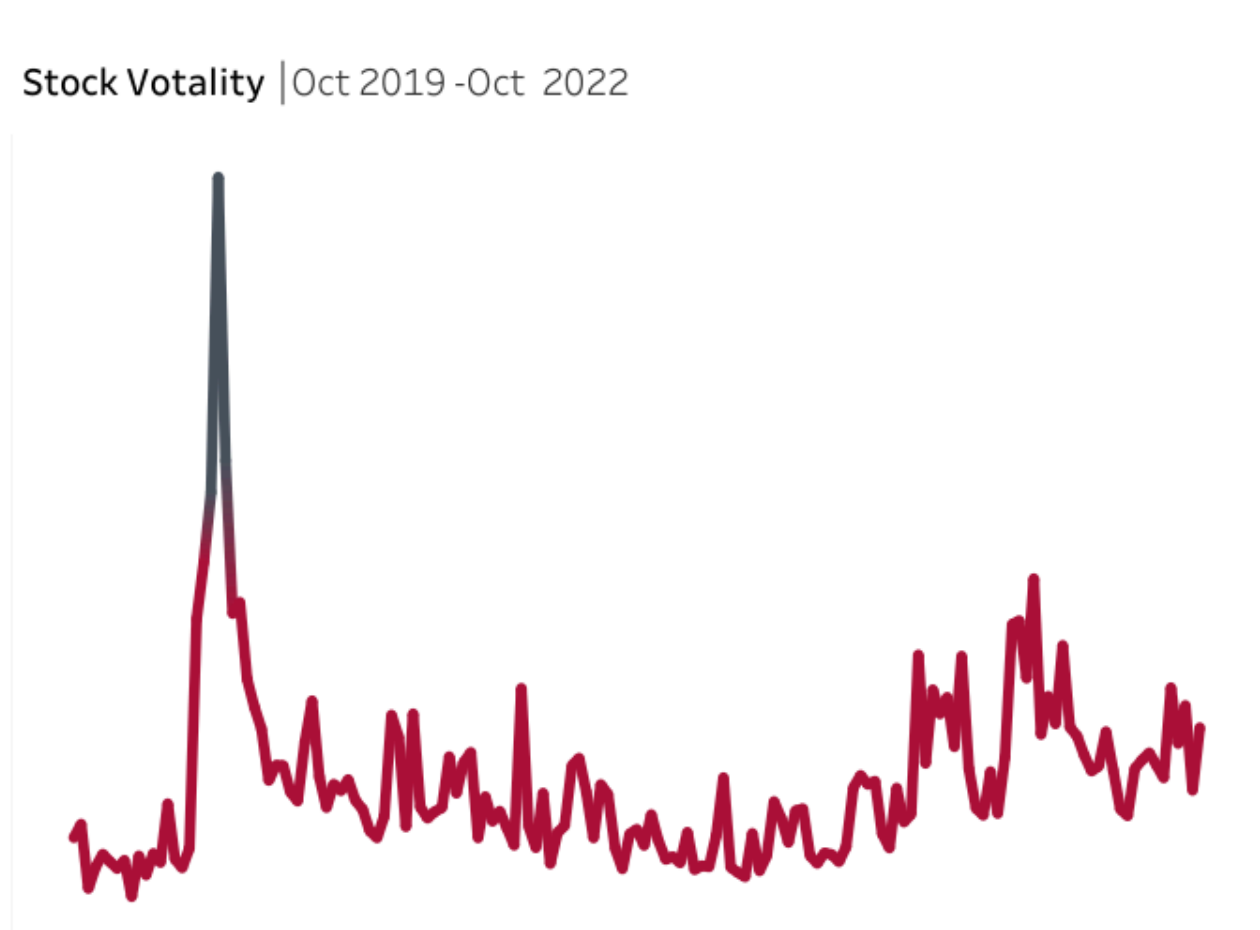
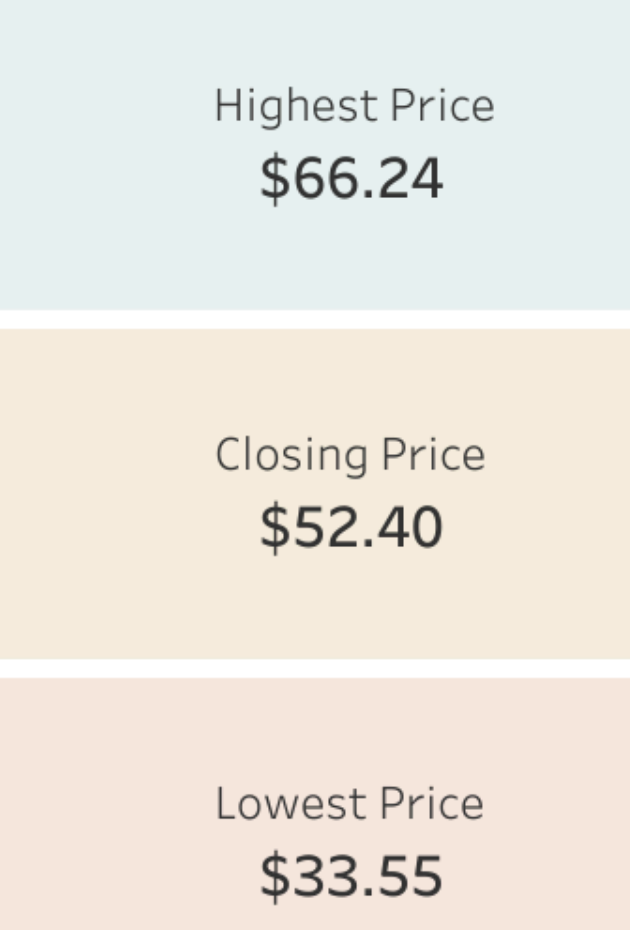
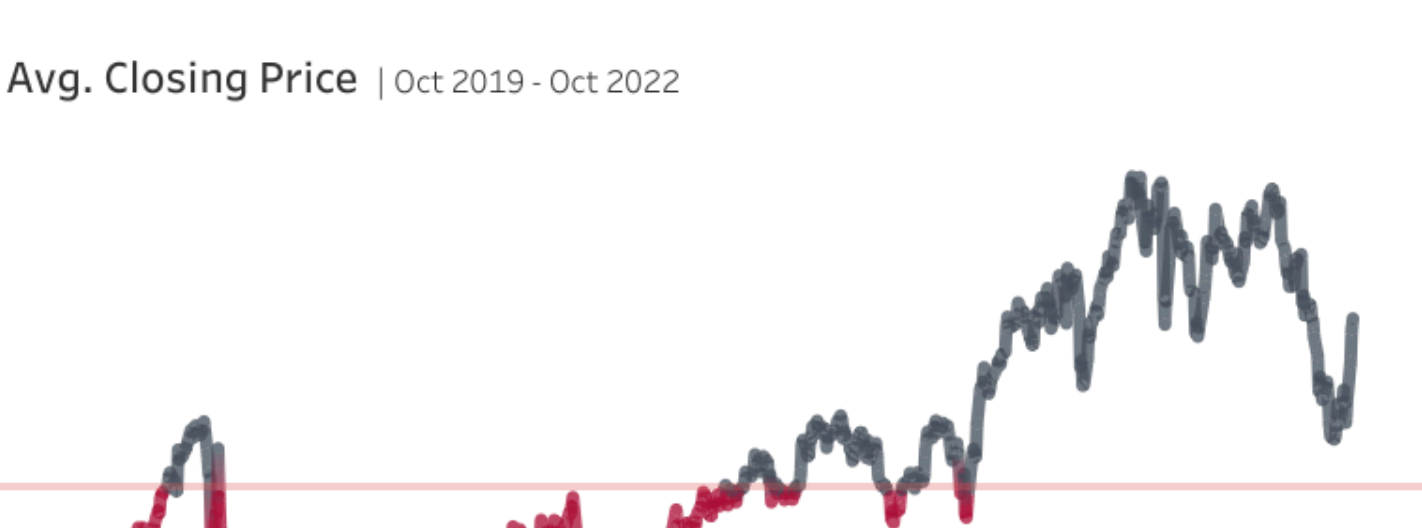
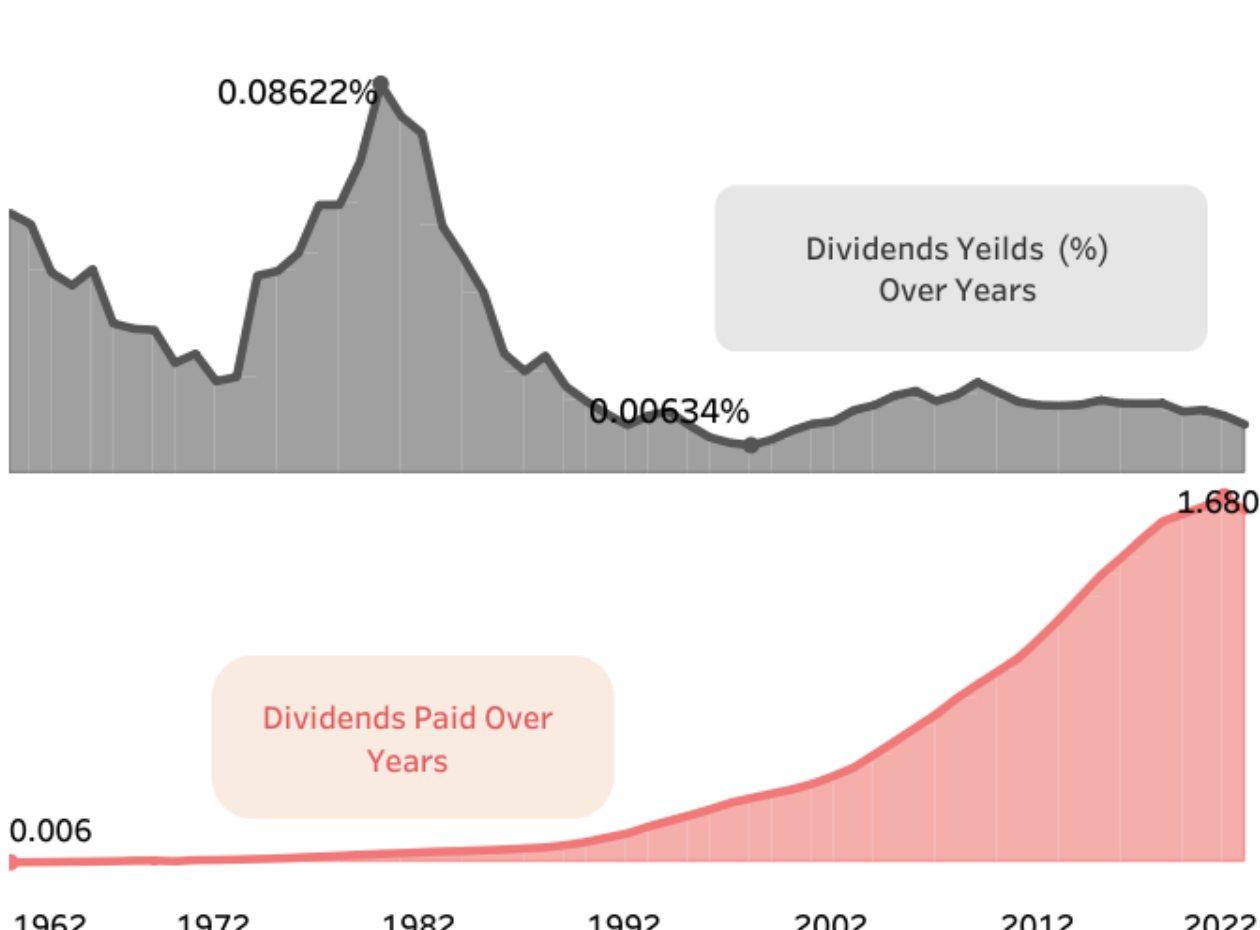
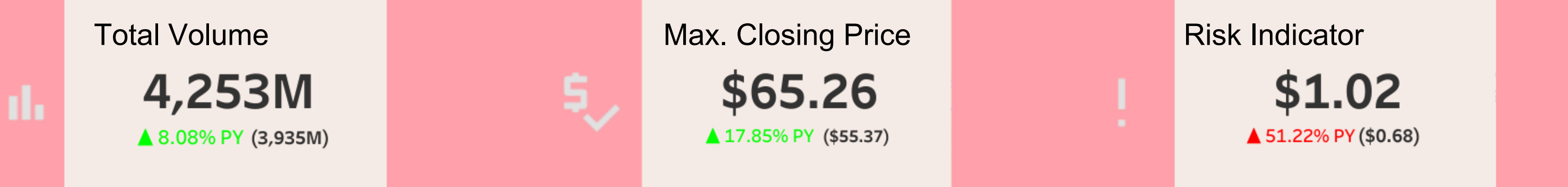
Area of Focus	Key Finding	Recommended Action
Channel Strategy	Uniform ROI across all Campaign Types.	Prioritize budget for high-revenue channels like Social Media.
Efficiency	Overall Weighted ROI stabilized at 6.01%.	Maintain current spend levels; focus on cost reduction.
Growth Opportunity	Significant drop-off at the top of the funnel (8.5% CTR).	A/B test ad creatives to improve Click-Through Rate.

The data confirms that the current marketing challenge is not a lack of scale or channel selection, but a bottleneck in engagement quality. We are successfully achieving massive reach and maintaining a high lead-to-sale conversion rate (21.94%), but the current ecosystem is **losing the vast majority of potential revenue at the Click-Through (CTR) stage**. Because our ROI remains stable over long durations without decay, we do not need more frequent campaign refreshes; rather, we need a higher creative hook to capture the billions of impressions currently being wasted. By focusing on Top-of-Funnel optimization and maintaining strict cost-caps during the Q2 period, we can maximize the efficiency of our \$62.3M budget and push the portfolio beyond its current 6.01% ROI ceiling.

Final Statement

The marketing portfolio is currently in a **Steady State** of performance. While risk is low due to consistent returns across all segments, the primary path to increased profitability is not through channel shifting, but through **optimizing the engagement funnel** to convert a higher percentage of our 9.1B impressions.

Coca.Cola (KO) Stock Market Performance | Oct 2019 - Oct 2022

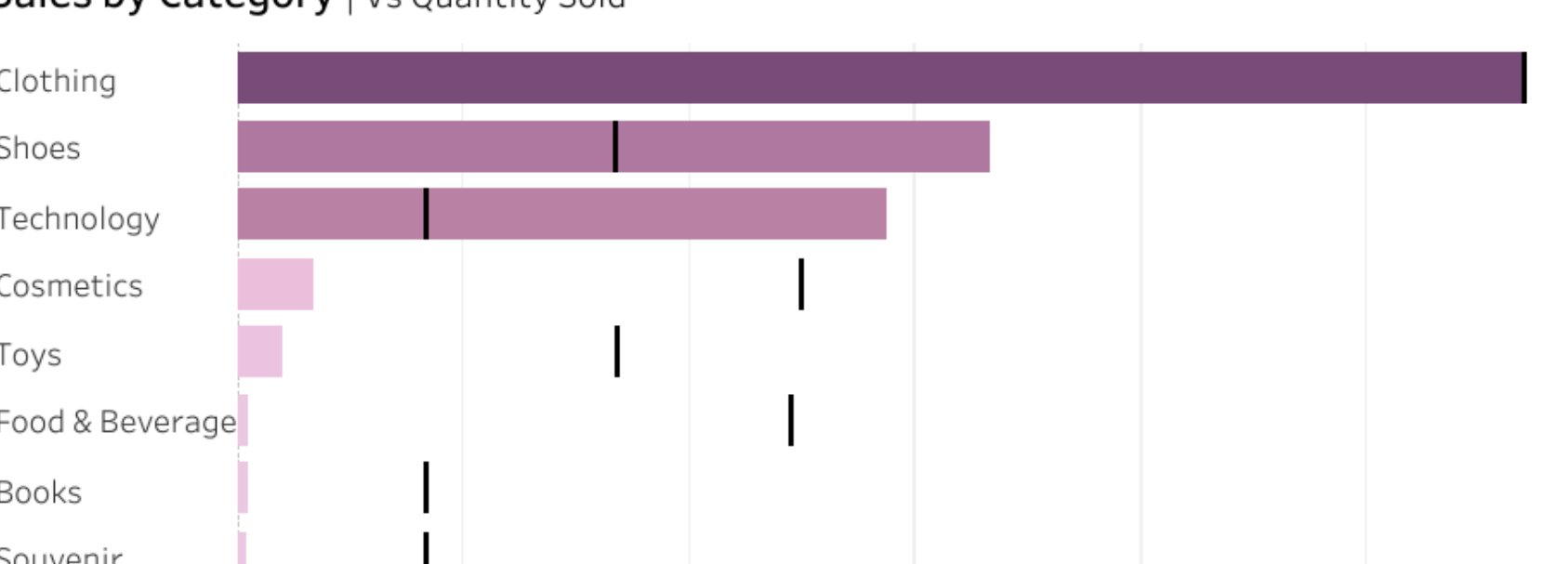


Sales Trends & Revenue Drivers

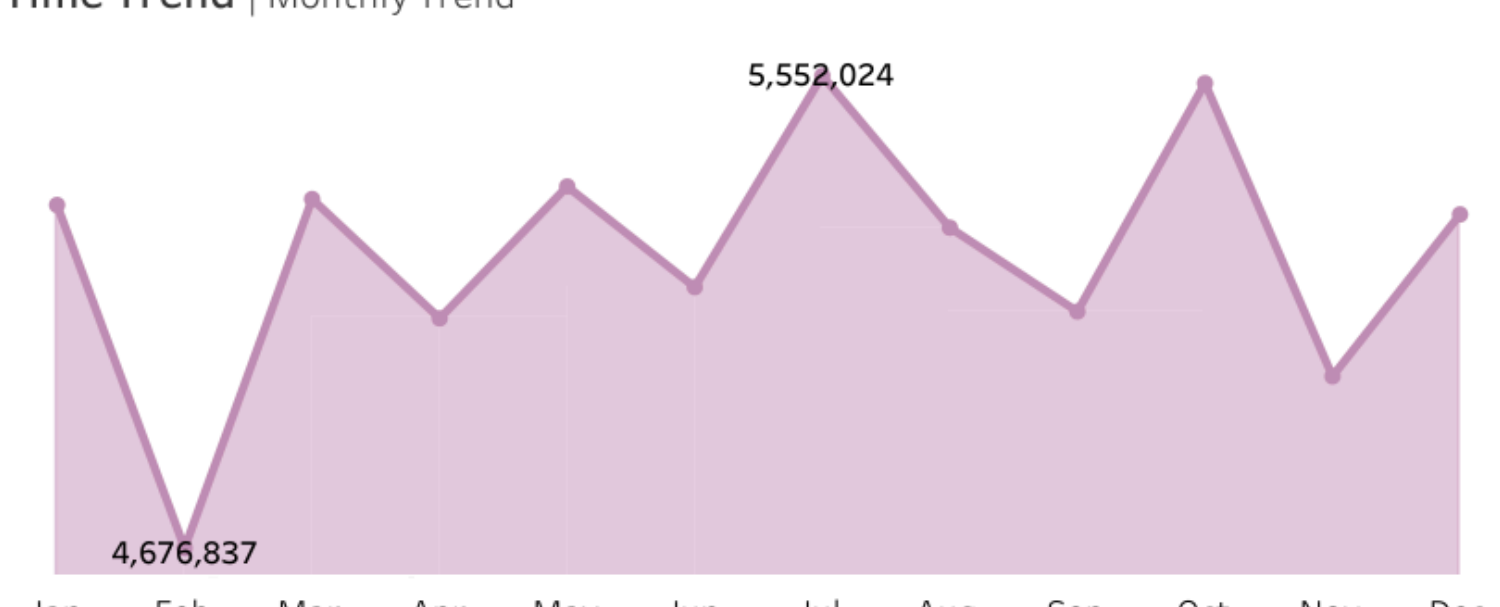
Exploring category-wise performance to highlight the backbone of total sales



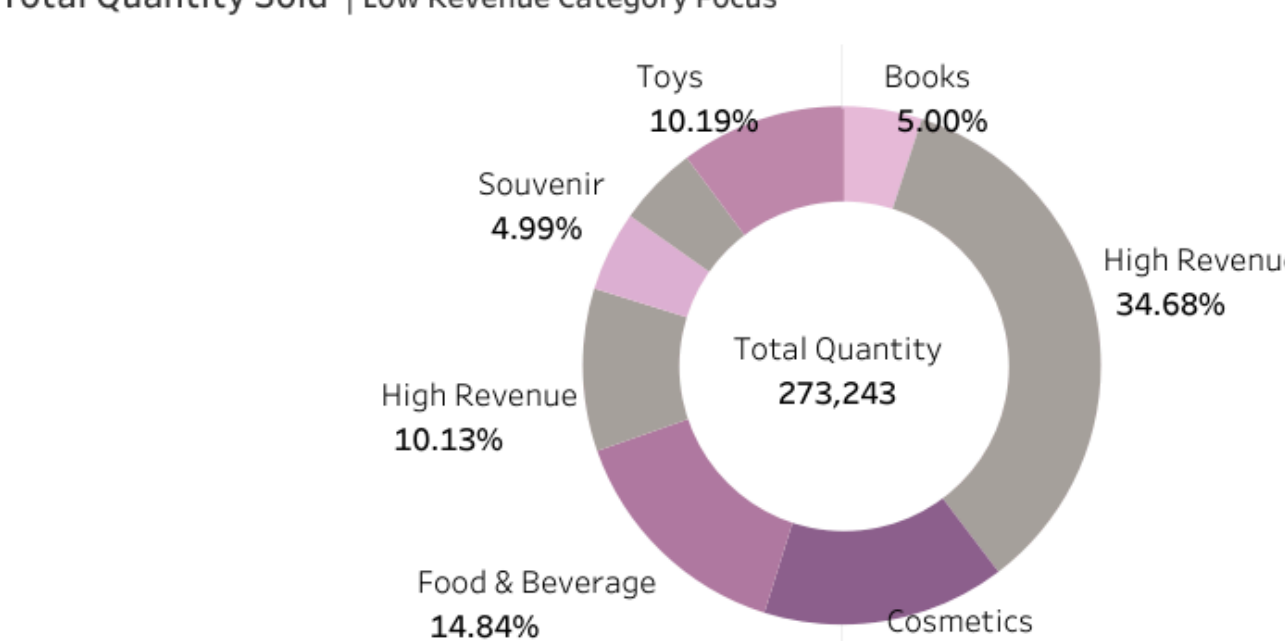
Sales by Category | Vs Quantity Sold



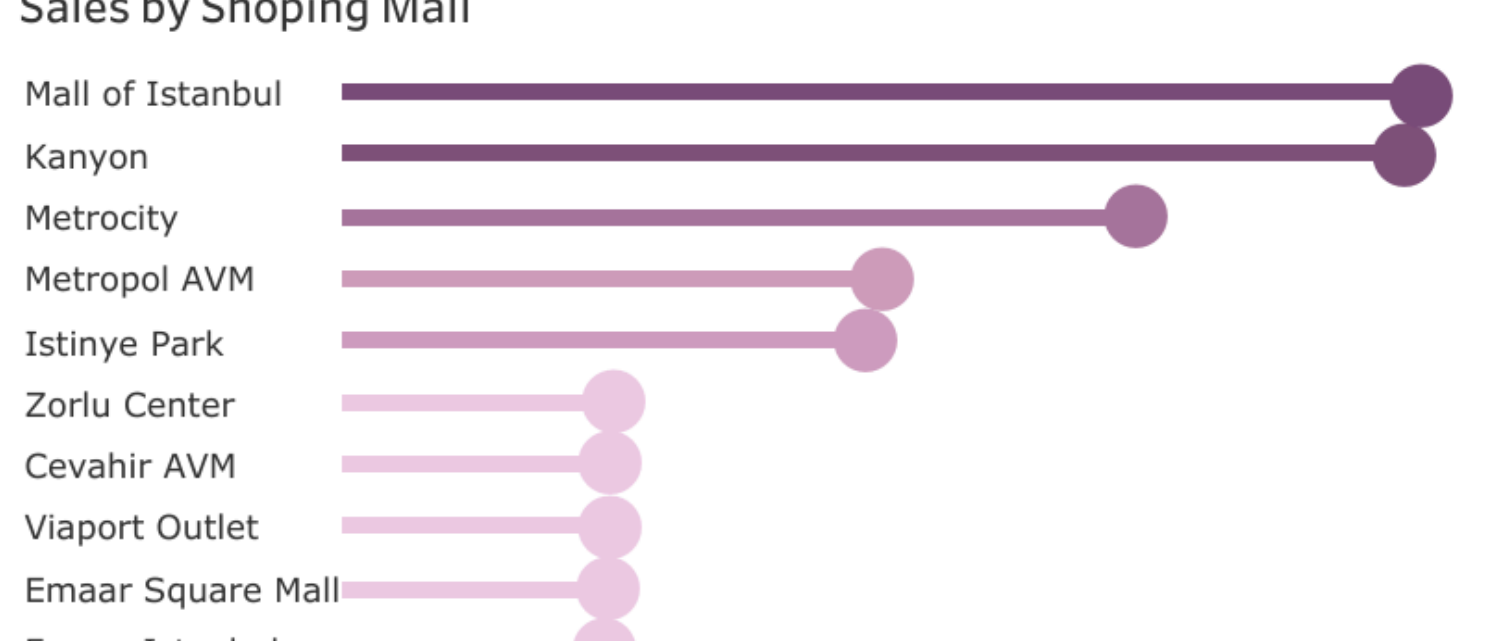
Time Trend | Monthly Trend



Total Quantity Sold | Low Revenue Category Focus

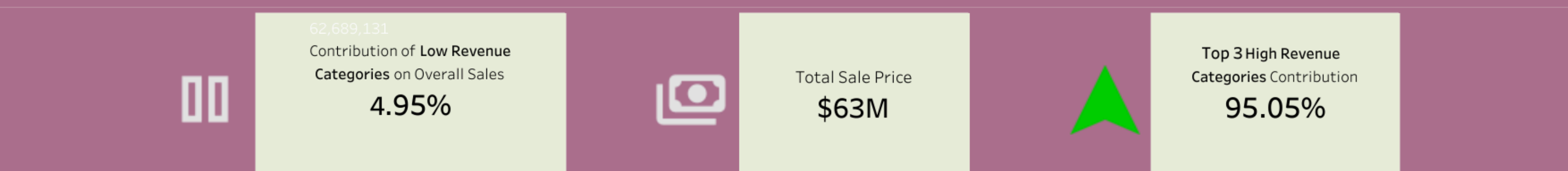


Sales by Shopping Mall

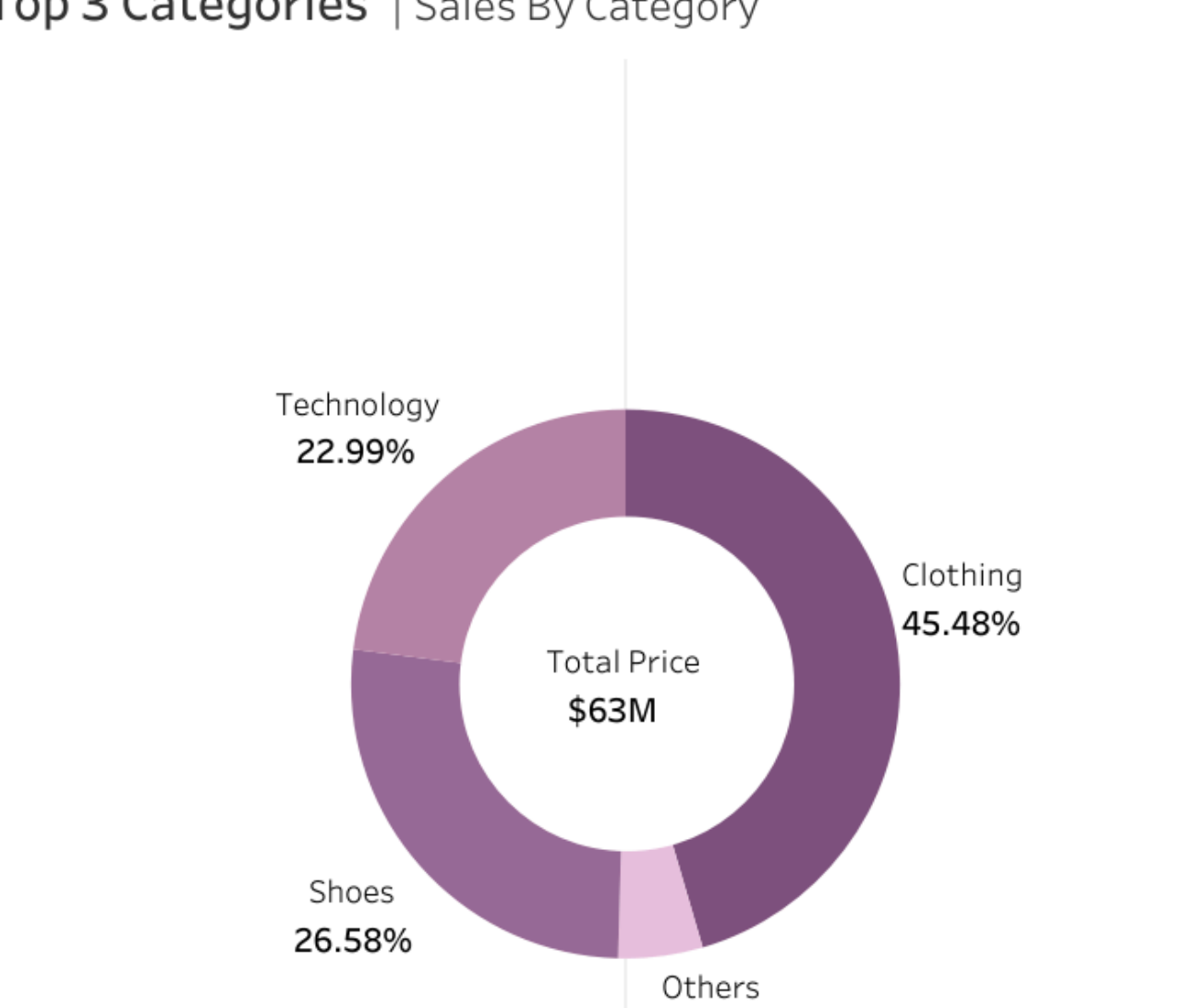


Sales Trends & Revenue Drivers

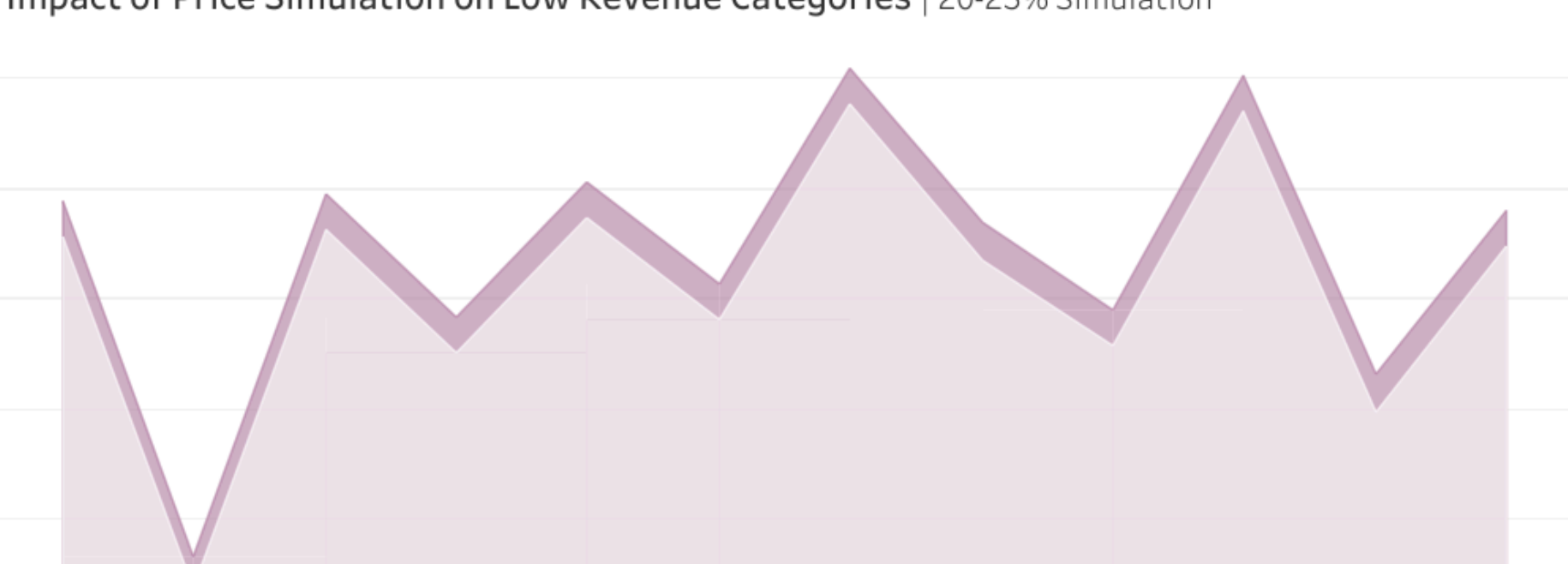
Exploring category-wise performance to highlight the backbone of total sales



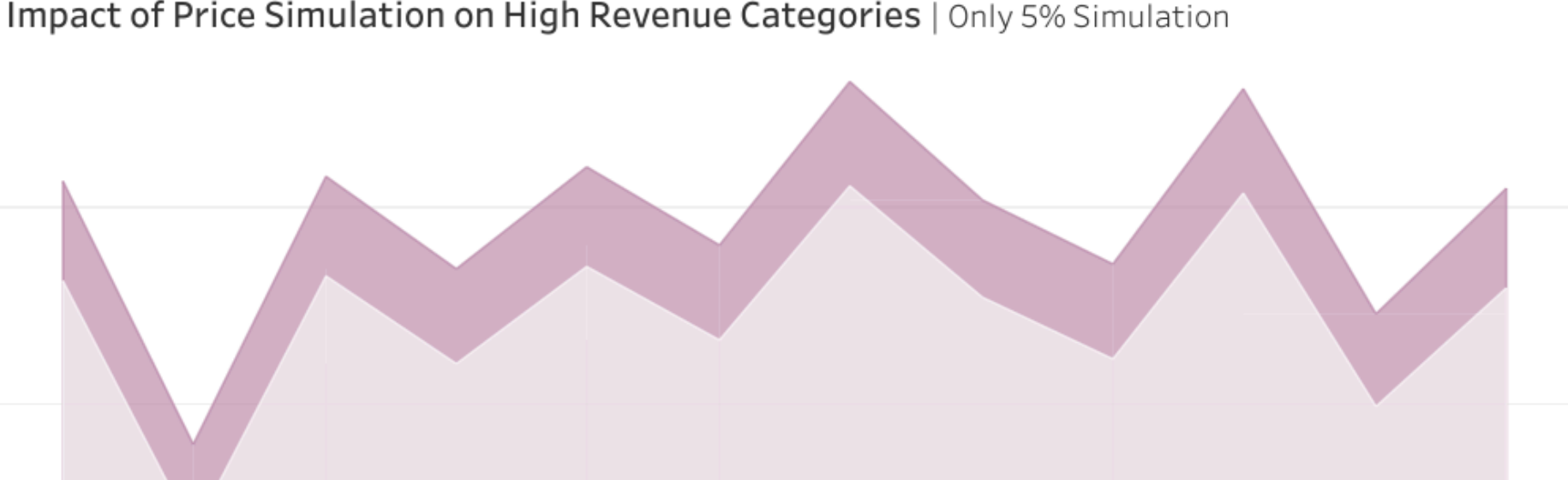
Top 3 Categories | Sales By Category



Impact of Price Simulation on Low Revenue Categories | 20-25% Simulation



Impact of Price Simulation on High Revenue Categories | Only 5% Simulation



Contact me

If you are looking for a detail-oriented data analyst, I would be happy to discuss how I can add value to your team

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